# UE23CS352B - Object Oriented Analysis & Design

## Mini Project Report

## Title: Path Planning Sandbox

***Submitted by:***

***Team Members***

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**Problem Statement:**

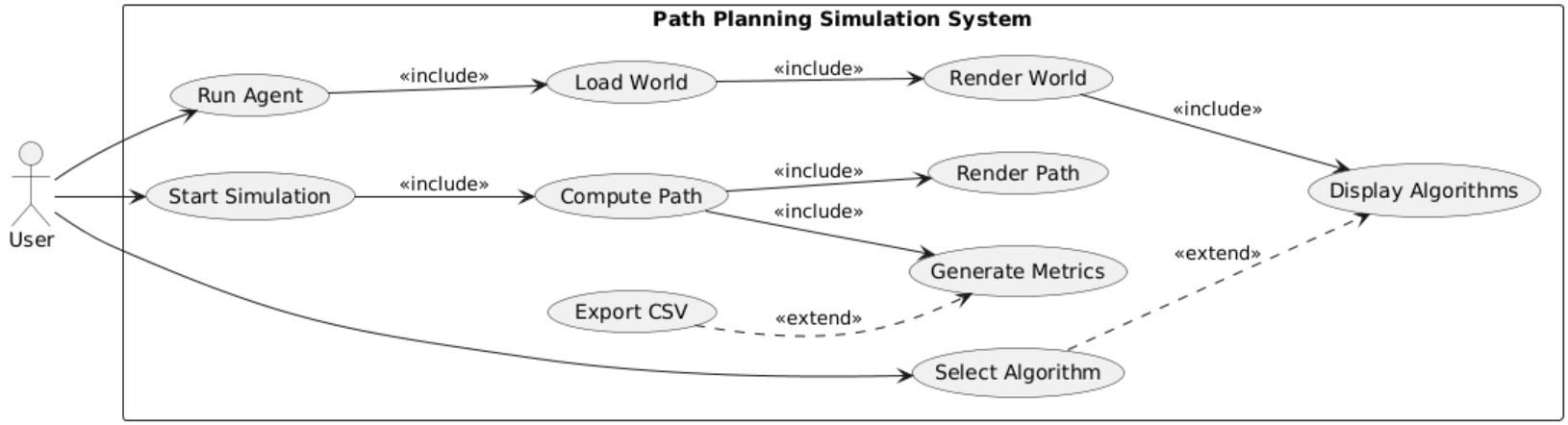
A modular, extensible system is needed to test, compare, and visualize different pathfinding algorithms across diverse map representations (2D grids and real-world geographic networks). The system must support dynamic algorithm selection, persistent execution history, and performance metrics tracking while maintaining clean separation of concerns through object-oriented design principles.

**Key Features:**

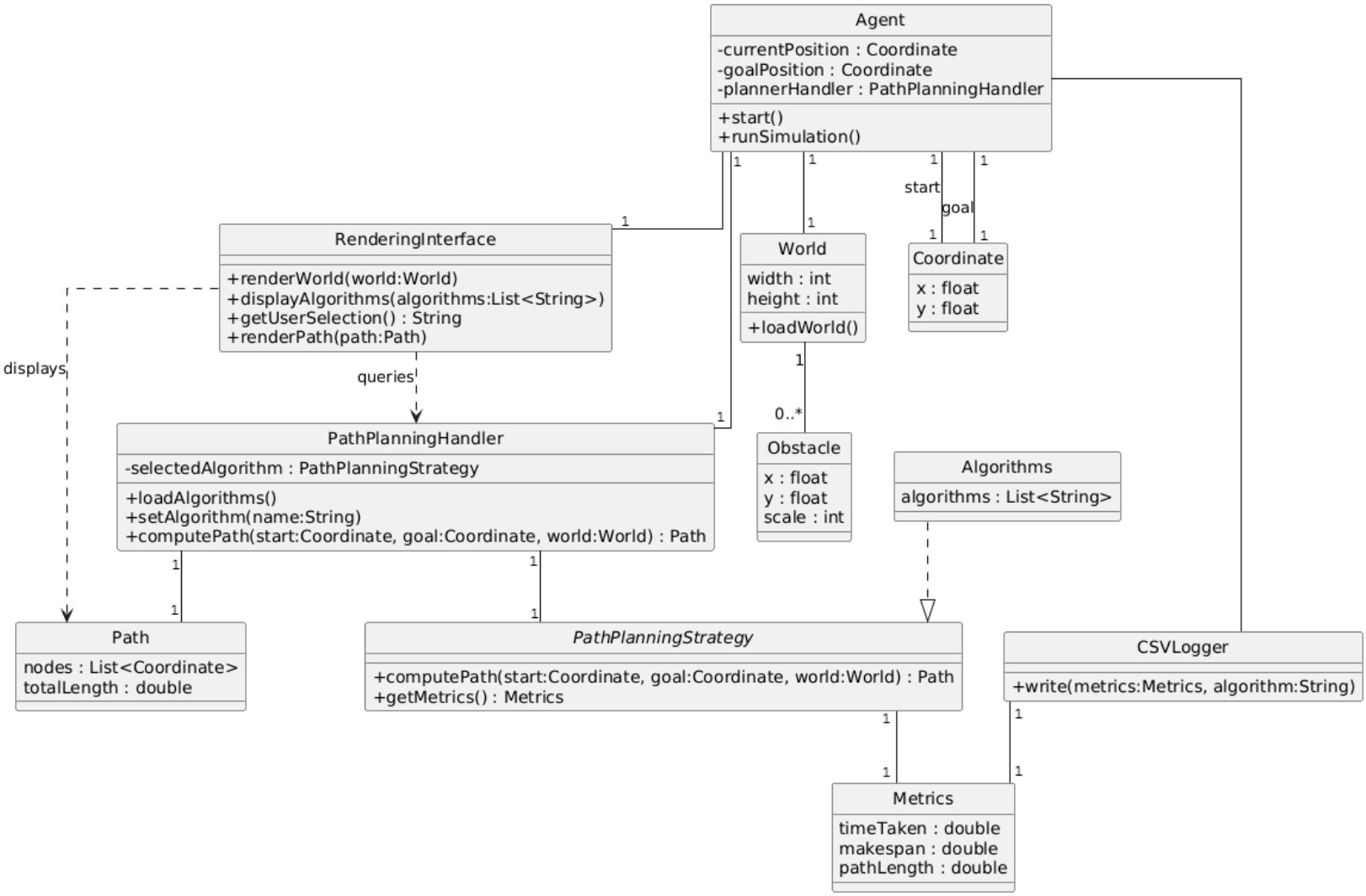
- Multiple pathfinding algorithms (Dijkstra's shortest path, Random Traversal baseline)
- Dual map support: JSON-based 2D grids and OpenStreetMap (OSM) XML networks
- Interactive web UI for coordinate input, algorithm selection, and result visualization
- H2 relational database for persistent execution history and performance logging
- Performance metrics tracking (execution time, path length, nodes expanded, success status)
- GeoJSON export for geographic pathfinding results
- Animated HTML visualization of traversed paths

Models:

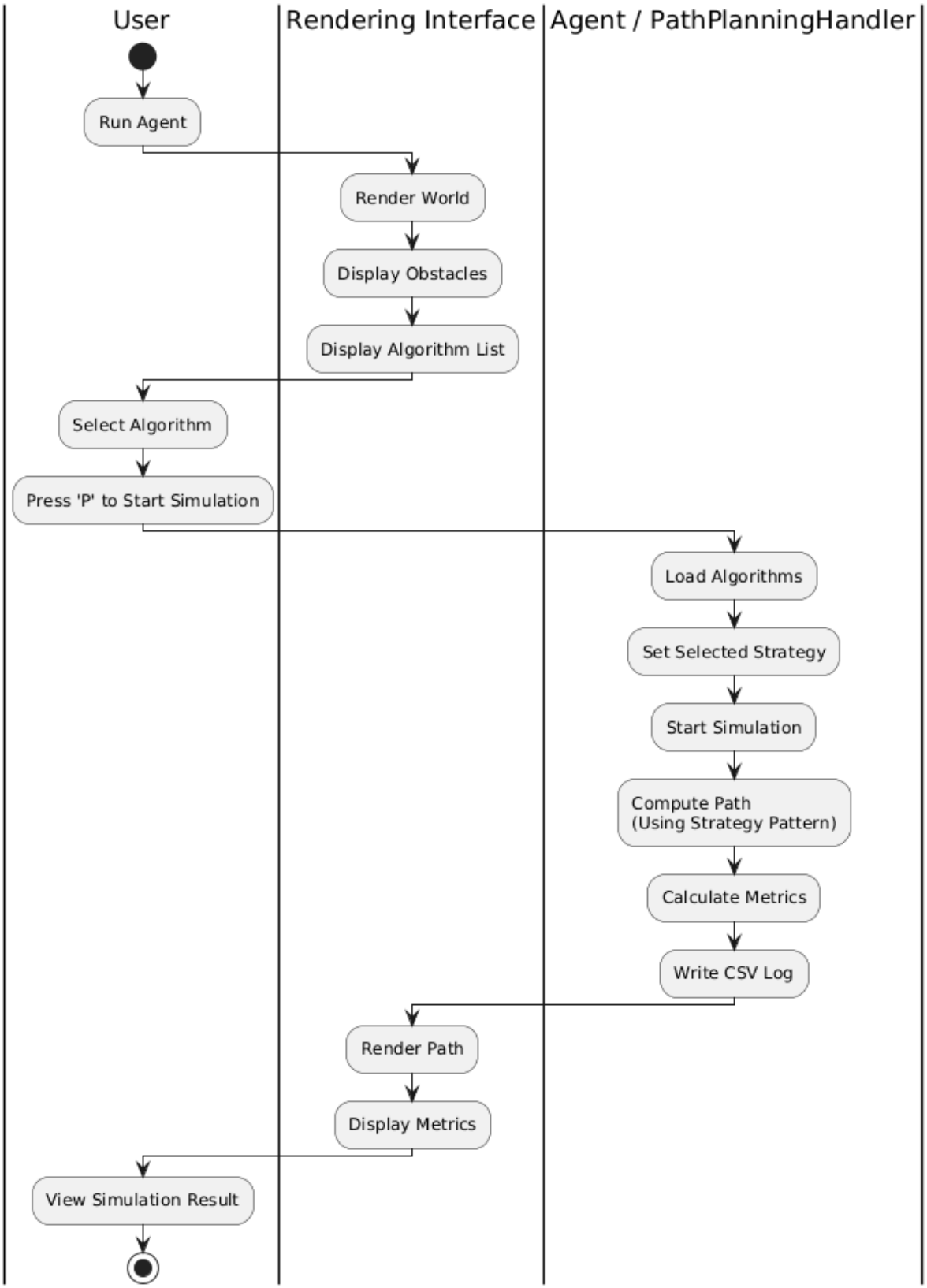
Use Case Diagram:



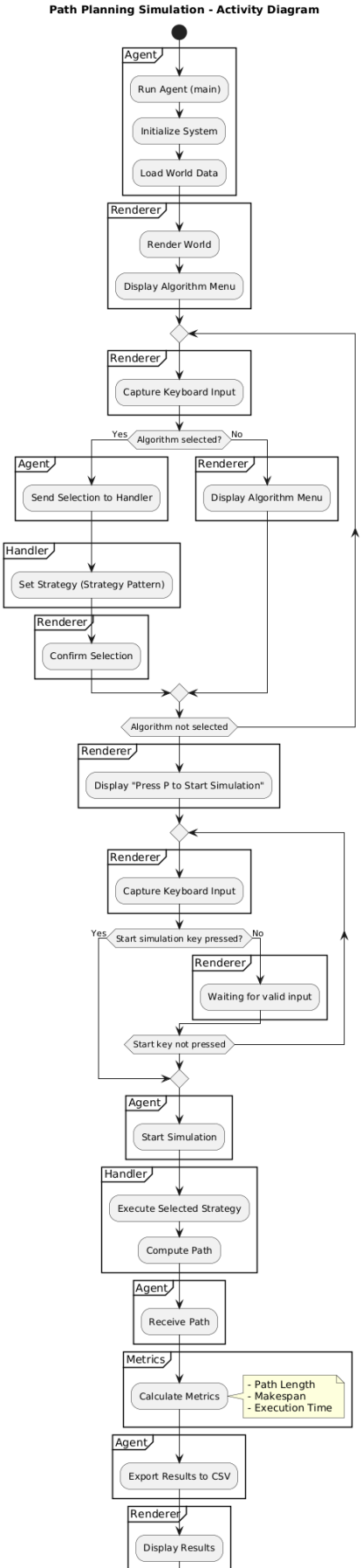
## Class Diagram:



## State Diagram:



# Activity Diagrams:



## Design Principles, and Design Patterns:

**MVC Architecture used?**
**Yes**

**Design Principles**
Separation of Concerns
The project is divided into 7 independent Maven modules, each handling a distinct responsibility:

- Core domain models are isolated in the core module
- Algorithm implementations reside in planners
- Input parsing logic is encapsulated in io
- Visualization is handled separately in rendering
- Metrics collection is independent in metrics
- This modularity prevents coupling and enables testing each layer independently

**Single Responsibility Principle (SRP)**
Each class has one reason to change:

- Coordinate - Manages position data only
- DijkstraPlanner - Implements Dijkstra algorithm only
- GridReader - Parses JSON grids only
- Renderer - Generates output visualizations only
- PathService - Orchestrates the workflow only

**Dependency Inversion**
The application depends on abstractions rather than concrete implementations:

- PathPlanner interface abstracts algorithm behavior
- GraphBuilder interface abstracts parsing logic
- Spring dependency injection inverts the dependency hierarchy, allowing loose coupling

**Open/Closed Principle (OCP)**
The system is open for extension but closed for modification:

- New pathfinding algorithms can be added by implementing PathPlanner without modifying existing code
- New map formats can be supported by creating new GraphBuilder implementations
- The core PathService remains unchanged when adding new planners

**Design Patterns**

1. Strategy Pattern (Behavioral)

Usage: PathPlanner interface with multiple implementations

- PathPlanner defines the interface for all pathfinding strategies
- DijkstraPlanner and RandomTraversalPlanner are concrete strategies
- PathService selects the appropriate strategy at runtime based on user input
- Allows runtime algorithm swapping without modifying service logic

2. Factory Pattern (Creational)

Usage: Map parsing and graph construction

- GridReader factory encapsulates JSON parsing complexity and returns AdjacencyGraph instances
- OsmGraphBuilder factory encapsulates OSM XML parsing complexity and returns AdjacencyGraph instances
- GraphBuilder interface abstracts the creation process
- Centralizes complex object creation logic and makes it maintainable

3. Adapter Pattern (Structural)

Usage: Coordinate abstraction for dual mapping systems

- Coordinate class acts as a universal adapter between two incompatible representations:
  - Cartesian integer coordinates (X, Y) for 2D grids
  - Double-precision Latitude/Longitude for geographic maps
- The core pathfinding engine remains agnostic to the underlying coordinate system
- Enables seamless switching between grid and OSM without algorithm modification

4. Singleton Pattern (Creational)

Usage: Spring Framework dependency injection

- PathService, Renderer, and database repositories are automatically managed as singletons by Spring
- Ensures single instances are shared across the entire application context
- Reduces memory overhead and maintains consistent state for shared resources

5. Repository Pattern (Behavioral)

Usage: Data persistence abstraction

- Spring JPA Repository interfaces abstract database operations
- PathExecutionLog entity is persisted through repository without exposing SQL queries
- Enables easy switching between database implementations

6. MVC Architecture

Usage: Complete separation of presentation, business logic, and data layers

- Model: Domain entities (Grid, Coordinate, PathResult), JPA entities

(PathExecutionLog)
- View: Thymeleaf templates (index.html, history.html, result.html)
- Controller: PathController receives HTTP requests and coordinates between services and views
- Clean routing: user input → controller → service → database/algorithms → view rendering

**Github link to the Codebase: (repository should be public)**

https://github.com/venator158/SandPP

## Screenshots

UI:

### SandPP - Path Planning Engine

#### Execute New Path

Map Input File (e.g. examples/grid1.json or examples/neighbour.osm)

examples/grid1.json

Start Coordinate (x,y or lat,lon)

0,0

Goal Coordinate (x,y or lat,lon)

4,4

Planner Algorithm

Dijkstra's Algorithm

Allow Diagonals (Grid Only)

Find Path

#### About SandPP

SandPP is an interactive path planning sandbox that allows you to visualize and evaluate different routing algorithms on Cartesian grids and OpenStreetMap data.

View Execution History

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### Path Execution Results

| Metric | Value |
|---|---|
| ID | 4 |
| Algorithm | dijkstra |
| Map | examples/neighbour.osm |
| Start | 12.975,77.670 |
| Goal | 12.978,77.676 |
| Execution Time | 1 ms |
| Nodes Expanded | 343 |
| Path Length Steps | 40 |
| Actual Distance | 0.008045564638526036 |

**Execution logged at:** 2026-04-20T12:08:05.355610

**View Rendering:**

Open Path in Geojson.io

Back to Menu | View Full History

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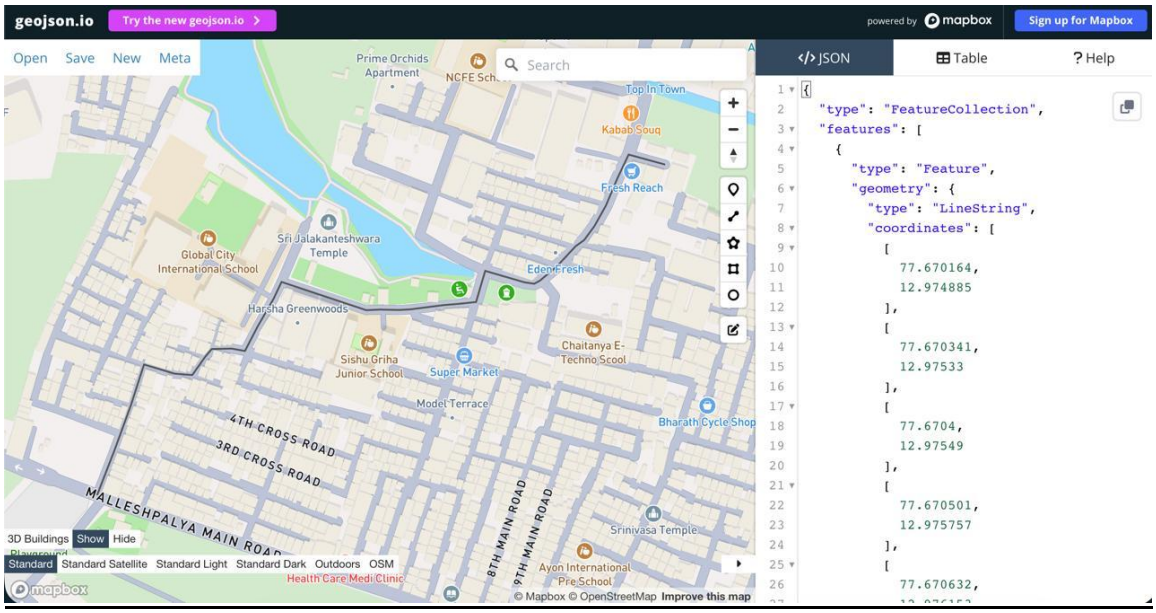


## Path Execution Results

| Metric | Value |
|---|---|
| ID | 5 |
| Algorithm | dijkstra |
| Map | examples/grid1.json |
| Start | 0,0 |
| Goal | 4,4 |
| Execution Time | 0 ms |
| Nodes Expanded | 19 |
| Path Length Steps | 9 |
| Actual Distance | 8.0 |

**Execution logged at:** 2026-04-20T12:11:13.119065

**View Rendering:**

The grid visualization was output directly to your local `output/index.html` folder.

For Web integration of Canvas rendering, this would require hosting the grid JSON as an API endpoint. But for now, you can open that file on your local machine.

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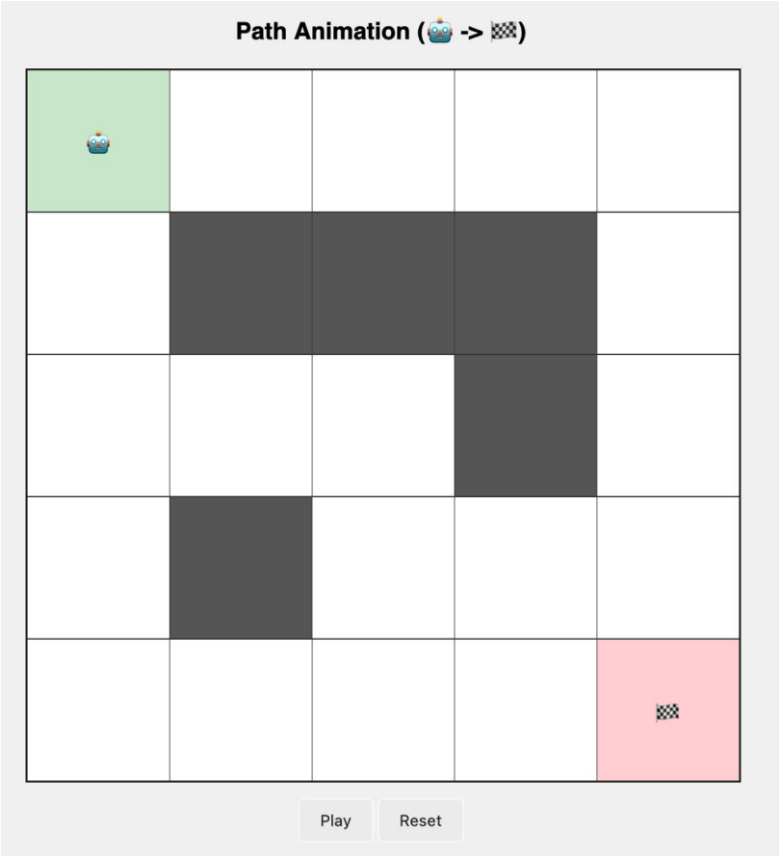
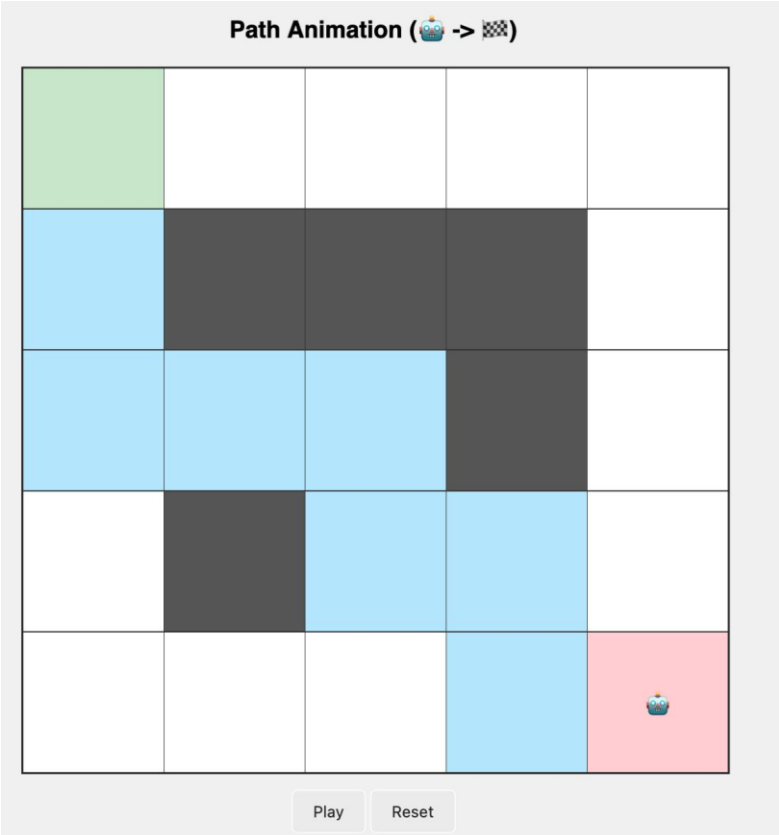

Path Animation (🤖 -> 🏁)
Play
Reset


Path Animation (🤖 -> 🏁)
Play
Reset

## Pathfinding Execution History

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| ID | Run Date | Map | Algorithm | Start → Goal | Length | Exec Time (ms) | Nodes Expanded | GeoURL |
|---|---|---|---|---|---|---|---|---|
| 1 | 2026-04-20T11:32:12.310311 | examples/grid1.json | dijkstra | 0,0->4,4 | 9 | 2 | 19 | Local |
| 2 | 2026-04-20T11:33:21.606949 | examples/grid1.json | dijkstra | 0,0->4,4 | 9 | 0 | 19 | Local |

Individual contributions of the team members:

| **Name** | **Module worked on** |
|---|---|
| **Unnathi Baskar** | Domain Analysis & Map/Environment Modeling |
| **Uday Gopan** | Algorithm Selection & Planning Execution |
| **Shreyas GS** | Result Representation, Visualization & Metrics |
| **Vishnu Prasath RS** | Persistence, Experiment Management & Reproducibility |